

NBA Career Longevity Prediction

A Strategic Data-Driven Approach

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Project Description & Objectives

Project Overview

- Based on the *UTS-ADVDSI NBA Career Prediction* challenge.
- **Goal:** Predict if an NBA rookie will play ≥ 5 years in the league using only Year 1 statistics.
- Strategic asset for Front Offices to optimize draft and renewal decisions.



Figure: 4 rookies at the beginning of the 2019-2020 season

Initial Objectives

- Build and evaluate binary classification models.
- Identify key features (PTS, MIN, etc.) correlating with longevity.
- Optimize **F1-score** to balance talent retention vs. financial risk.
- Handle class imbalance and ensure model generalization.

The Dataset: Features & Structure

- **Source:** UTS-ADVDSI 22-02 NBA Career Prediction (Kaggle).
- **Data Scope:** Performance statistics of NBA players during their rookie season.
- **Dataset Size:**
 - Training and test set: 8,000 samples.
 - Number of features: 19 performance metrics.
- **Target Variable (`target_5yrs`):**
 - **1:** Career longevity ≥ 5 years.
 - **0:** Career longevity < 5 years.

Statistics Captured

Games Played (GP), Minutes Played (MIN), Points (PTS), Field Goals (FGM/A, FG%), 3 Points (3PM/A, 3P%), Free Throws (FTM/A, FT%), Rebounds (OREB, DREB, REB), Assists (AST), Steals (STL), Blocks (BLK), and Turnovers (TOV).

Data Preprocessing: Error Detection & Cleaning

An initial inspection (`df.describe()` and boxplots) revealed impossible data and extreme outliers.

1. Impossible Data Errors:

- **GP:** Max games allowed is 110. Values exceeding this are data entry errors.
- **Percentages:** FG%, 3P%, and FT% are mathematically constrained to $\leq 100\%$.
- **Negative Values:** PTS, REB, AST, BLK, and TOV cannot be negative.

2. Statistical Outliers:

- Presence of extreme anomalies that skew model training (especially linear models).
- Solution: Filtering the dataset by setting reasonable maximums based on historical NBA rookie records.

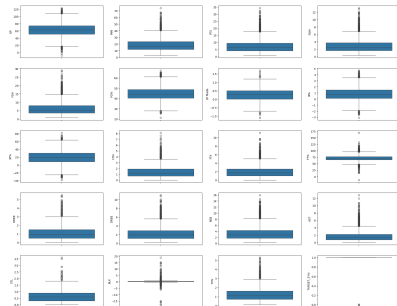
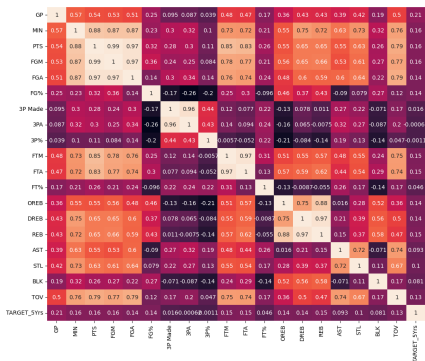


Figure: Initial boxplot distribution showing outliers and range errors.

Correlation Analysis & Feature Engineering



Key Correlation Insights:

- **Multifactorial Nature:** No single variable has a strong correlation with the target; longevity is a complex problem.
- **High Signal Features:** GP and MIN show the highest correlation, suggesting that trust and opportunity are the most robust signals for survival.
- **Negligible Factors:** 3P shooting metrics show near-zero correlation, likely due to the dataset era.

Simplification Strategy (Reducing Multicollinearity):

- **Offensive Volume:** Removed FGM and FGA; retained PTS (total production) and FG% (efficiency).
- **Redundant Volume:** Removed 3PA, FTA, and DREB due to near-perfect correlation with made shots/total rebounds.
- **Specialized Skill:** Retained OREB as it represents a high-effort skill distinct from total rebounds.

Figure: Heatmap of the Correlation Matrix.

EDA: Distribution of Playing Time and Scoring

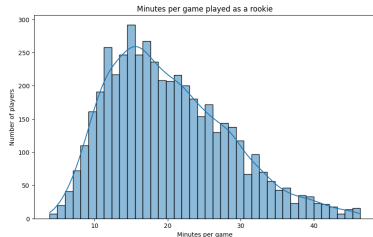
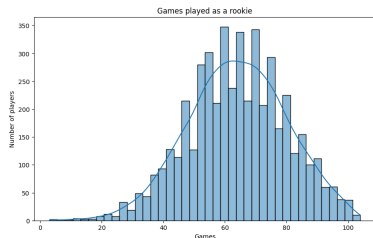


Figure: Distributions of GP and MIN.

Development & Exposure (GP):

- High appearances (GP centered 60-70).
- Necessary for the development of the player
- Big sample size for evaluate the player

The Bench Role Reality (MIN & PTS):

- Rotational reserves (MIN peaks 15-20).
- Skill Gap: Initial lack of physical maturity.
- Learning Curve: Not adapted to NBA speed.

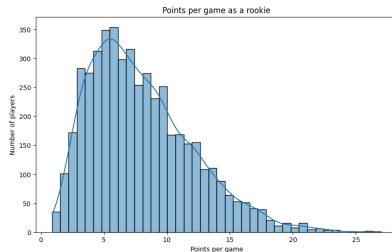


Figure: Distribution of Points per game.

Class Imbalance and Performance Correlation

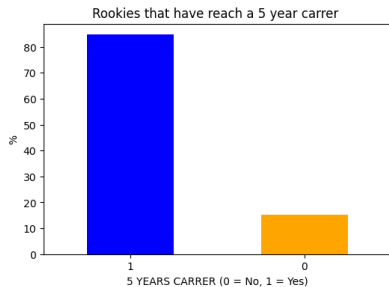


Figure: Distribution of Career Longevity (Target Variable).

The Challenge of Class Imbalance:

- **Dominant Class (1):** Approximately 85% of observations.
- **Minority Class (0):** Only about 15% of observations.
- **Context:** Reflects the high barrier to entry; NBA players are already the top 0.01% of talent worldwide.
- **Expectation:** Longevity is the statistically expected outcome for vetted prospects.

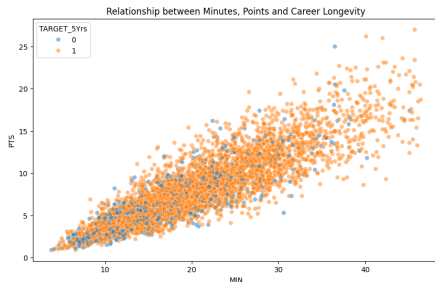


Figure: Relationship: Minutes and Points by Target.

Relationship between Minutes, Points, and Longevity:

- **Success Zone:** High concentration of Class 1 (orange) in the upper-right section.
- **Threshold:** Most players with > 25 MIN and > 10 PTS reach the 5-year mark.
- **Risk zone:** Longevity is at high risk if a player cannot earn $> 10 - 12$ minutes of playing time, regardless of efficiency.

Model Development: Algorithms and Hyperparameter Tuning

To identify the most effective predictive engine, a systematic evaluation of six different classification algorithms was conducted using **GridSearchCV** for hyperparameter optimization.

Tested Algorithms & Parameters:

- **Decision Tree:** `max_depth` (1-20), `class_weight` (None,Balanced).
- **Gaussian Naive Bayes:** `var_smoothing` (log scale).
- **Linear Perceptron:** `early_stopping` (true/false), `class_weight` (None,Balanced).
- **K Nearest Neighbor:** `n_neighbors` (1-7).
- **AdaBoost:** `n_estimators` (10-50), `learning_rate`.
- **Random Forest:** `max_depth` (4-10), `n_estimators` (10-60).

Evaluation Metrics:

- Accuracy
- Precision (Macro & Weighted)
- Recall (Macro)
- **F1-Score (Macro & Weighted)**

Priority Metric

F1-Macro was selected as the primary scoring metric to ensure a fair balance between detecting both successful careers and potential "busts" despite the class imbalance.

Detailed Performance Matrix: Why GNB?

Performance Matrix by Winning Metric:

Model (Target Metric)	TP	TN	FP	FN	F1-M
Decision Tree (Acc)	1220	0	221	0	0.4583
L. Perceptron (Prec M)	1218	1	220	2	0.4581
Decision Tree (Prec W)	612	157	64	608	0.5431
Decision Tree (Rec M)	549	169	52	671	0.5428
Gaussian NB (F1-M)	886	104	117	334	0.5564
AdaBoost (F1-W)	1194	12	209	26	0.4913

Best model for f1_macro: Gaussian Naive Bayes (Value: 0.5564)

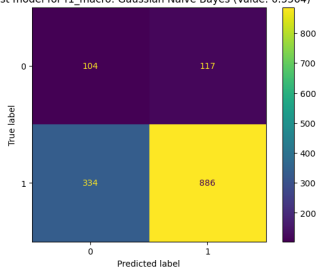


Figure: Selected GNB Model Confusion Matrix (F1-Macro optimized).

The Accuracy Paradox:

- Models optimized for accuracy (e.g., Decision Tree) tend to default to the majority class (TN=0).
- This creates a "silent failure" as the model becomes blind to the minority class.

Alternative Decision Trees (Prec W / Rec M):

- These models succeed in identifying over 50% of the minority class (TN=157 and TN=169).
- However, they fail critically in Class 1, producing an excessive number of False Negatives (FN=608 and FN=671).

Strategic Balance of GNB:

- Optimal Trade-off:** GNB is the only model that effectively captures Class 0 signals (TN=104) without collapsing the predictions for Class 1.
- Macro F1 Priority:** Provides the most honest and functional scouting report for decision-making.

Alternative Strategies: Support Vector Machine

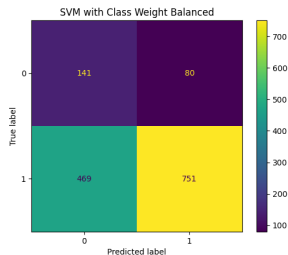


Figure: Confusion Matrix for Optimized SVM.

Model Architecture

```
svm_model = SVC(kernel='rbf',  
class_weight='balanced', probability=True)
```

Addressing the 85/15 Data Imbalance:

- **Cost-Sensitive Learning:** The `class_weight='balanced'` parameter is crucial. It penalizes Class 0 errors more heavily to counteract the majority bias.
- **RBF Kernel:** Captures non-linear feature relationships, projecting data into higher dimensions for better separability.
- **Decision Boundary:** Maximizes the margin between classes, providing a rigid boundary that differs from GNB's probabilistic overlap.

Performance Metrics:

- **F1-Macro Score: 0.54.**
- **Class 0 Specificity:** The model prioritizes Precision in Class 0 (141 TN).
- **Predictive Behavior:** It acts as a strict filter; it is designed to reduce False Positives in the minority class, ensuring that "Bust" predictions are highly reliable.

Advanced Techniques: SMOTE

What is SMOTE and why use it?

- **Oversampling Strategy:** SMOTE creates synthetic examples to balance the 85/15 class ratio.
- **Linear Interpolation:** It generates new points along the lines connecting minority samples (Class 0) and their k -nearest neighbors.
- **Generalization:** By expanding the minority region, it helps the model learn a broader decision boundary rather than memorizing noise.

SMOTE Results (GridSearch Optimization):

Model (SMOTE)	TP	TN	FP	FN	F1-M
AdaBoost (Acc)	1144	22	199	76	0.4925
Gaussian NB (All)	610	158	63	610	0.5552
Random Forest (F1-M)	1019	58	163	201	0.5451

The Performance Hierarchy:

- **Balanced SVM ("King of Recall"):** Found 141 failures but is too pessimistic, discarding 469 good players.
- **Original GNB (non-SMOTE):** Found 104 failures while discarding only 334 good players.
- **SMOTE Random Forest:** Only found 58 failures. Synthetic data did not necessarily help the trees find a better boundary in this dataset.

Selection: Random Forest (SMOTE)

- The Random Forest was the most **stable** model when handling synthetic clusters.
- **Success Preservation:** It maintained high performance on the majority class (1019 TP) while attempting to learn minority patterns.
- **Distribution :** Tree-based models navigate the artificial SMOTE space effectively because they do not assume a normal distribution (as GNB).

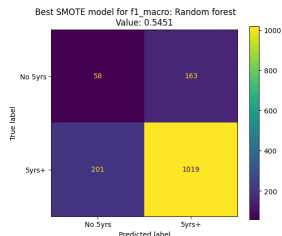


Figure: Confusion Matrix: SMOTE Random Forest.

Neural Networks

Strategy A: MLP with SMOTE (Balanced)

- **Architecture (16, 8):** A "funnel" structure with ReLU activation and Adam optimizer to capture non-linear patterns.
- **Result:** F1-Macro of 0.5352.
- **Behavior:** Prioritizes the minority class (Busts), reaching a "performance plateau" similar to SVM.
- Corrects Class 0 bias but mistakenly discards 3 successful players for every bust identified.

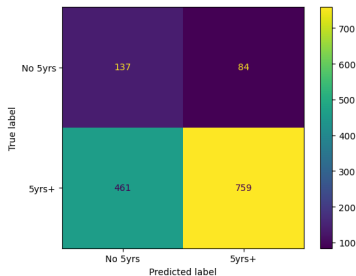


Figure: Confusion Matrix: MLP + SMOTE.

Strategy B: MLP without SMOTE (Imbalanced)

- The same architecture was trained on the original 85/15 dataset to evaluate balancing necessity.
- **Lazy Strategy:** Without balancing, the network converges toward the majority class.
- While overall Accuracy is high, the Recall for Class 0 drops to near-zero.
- **Result:** F1-Macro Score of 0.4706

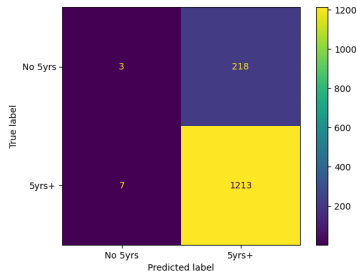


Figure: Confusion Matrix: MLP Imbalanced.

Note: Complex models like MLP require data balancing to avoid becoming simple majority-class predictors.

Ensemble Methods: Seeking the "Wisdom of the Crowd"

Strategy: Combining the "Individual Experts"

- After hitting a consistent F1-score ceiling of 0.55, the top models (**RF**, **GNB**, **SVM**) were merged into ensembles to cancel out specific individual errors.
- Soft Voting** : Averaging the probability (confidence level) of each model. A highly confident model can outvote others that are uncertain.
- Stacking**: A two-layer structure where a (Logistic Regression) learns which expert to trust in specific situations to correct final outputs.

Predictive Stalemate: The Data Ceiling

- Low Score**: Both models failed to break the barrier, converging to F1 scores of **0.5157** and **0.5311**.
- Weak Signal**: First-year rookie stats do not contain enough information to isolate career longevity.
- External Noise**: Career drivers like injuries, work ethic, and coaching changes are variables the current dataset simply cannot capture.

Opinion: *"The signal for a 5-year career is not fully formed in Year 1. Longevity is revealed in Years 2-3 through development and adaptation."*

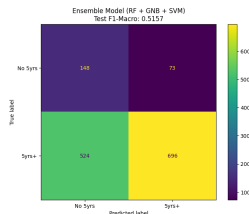


Figure: Confusion matrix: Soft Voting.

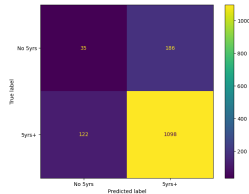


Figure: Confusion matrix: Stacking.

Targeted Feature Selection: Top-5 Strategy

Reducing Noise: The Focused Selection

- To mitigate bias and noise, the model transitioned from the full dataset to the five features most correlated with career longevity.
- **GP & MIN (Opportunity)**: A rookie playing most of the season has already earned professional coaching trust.
- **PTS & FG% (Offensive Dominance)**: Scoring efficiency is the primary "gate" to long-term NBA contracts.
- **REB (Physical Presence)**: Essential for role-players to sustain careers beyond pure scoring.

GridSearch Results (Focused Features):

Model	Score	TP	TN	FP	FN	F1-M
GNB (Top 5)	F1-M	1064	103	118	114	0.5660
RF (Top 5)	F1-M	1019	58	163	201	0.5451
SVM (Top 5)	Recall	711	141	80	469	0.5367
MLP (Top 5)	Acc	1111	35	186	109	0.5184

Results: Optimized GNB

- **Statistical Superiority**: Achieves the project's highest **F1-Macro (0.5660)**, outperforming every complex algorithm.
- **Aggressive Bias**: The ideal tool for high-budget franchises that prioritize talent discovery over risk mitigation.
- **The Trade-off**: Only flags 25% of Class 0. It is a superior scout for potential stars, but a poor filter for risk.

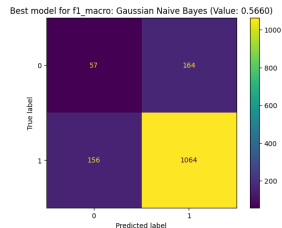


Figure: Confusion Matrix: GNB Top-5 Features.

Final Results: Comparison of Selected Models

Project Selection: The Top Performing Architectures

- Three distinct strategies were selected based on their capacity to balance talent discovery and risk management.
- **GNB Top 5:** The most efficient model, focusing on the five most impactful features.
- **GNB General:** A balanced model using the full statistical spectrum of the rookie year.
- **SVM (Balanced):** A risk-averse model designed to maximize the detection of potential career failures.

Final Performance Table (Test Set):

Model Strategy	F1-Macro	TP	TN	FP	FN
Top 5 (var_smoothing=0.001)	0.5660	1064	57	164	156
GNB General (var_smoothing=0.001)	0.5564	886	104	334	117
SVM (Balanced Class Weights)	0.54	751	141	459	80

Strategic Interpretation:

- **Maximum Efficiency:** The GNB Top 5 yields the highest F1-Macro, proving that reducing dimensionality to core "opportunity" metrics (MIN, GP, PTS) optimizes career prediction.
- **The Talent Tax:** While the SVM finds the most failures (141 TN), it also discards 469 potentially successful players.
- **The Performance Ceiling:** The small margin between models (0.54 – 0.56) confirms that first-year data has a natural information limit for long-term career projection.

Final Conclusions

The "No One Fits All" Paradigm

The choice of algorithm is not a purely mathematical decision; it depends entirely on a **franchise's financial health** and its competitive window.

- **Aggressive Rebuilding (GNB Top-5):** The optimal choice for teams seeking future stars at any cost. It prioritizes talent discovery with a high tolerance for False Positives.
- **Balanced Growth (Full-Feature GNB):** A bridge for teams with a salary cap cushion. It considers the full spectrum of rookie data to find "diamonds in the rough" without being as extreme as the Top-5 model.
- **Contenders & Risk Mitigation (SVM):** Best for championship-level teams where signing a "bust" is a costly distraction. It acts as a conservative filter to protect the roster.

The Rookie Information Ceiling

- The consistent **F1-Macro barrier (0.56)** across all complex models proves that the "signal" for a 5-year career is not fully formed in Year 1.
- Crucial drivers of longevity, such as player work ethic, injuries, coaching changes, and team chemistry, are variables that raw box scores simply cannot capture.